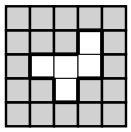
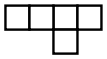
Preview: y2003_q18 (v1)

Covering a shape with pieces of squares: count the squares, then turn the pieces

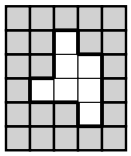
Part A: Each question is worth 3 points

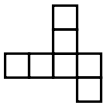
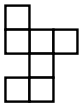
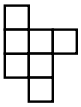
1. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



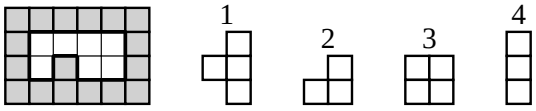
A. ***B.*** ***C.***  ***D.*** ***E.***

2. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



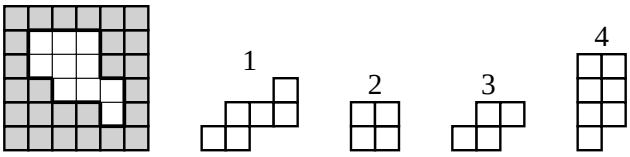
A. ***B.***  ***C.***  ***D.***  ***E.***

3. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2 ***B.*** 1+4 ***C.*** 2+3 ***D.*** 2+4 ***E.*** 3+4

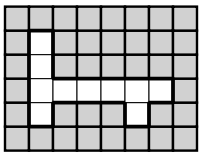
4. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.

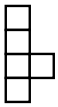
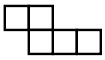


A. 1+2 ***B.*** 1+3 ***C.*** 1+4 ***D.*** 2+3 ***E.*** 2+4

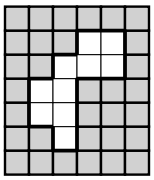
Part B: Each question is worth 4 points

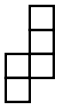
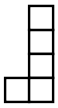
5. Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.



A.  ***B.*** ***C.*** ***D.***  ***E.***

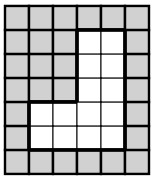
6. Two pieces that are exactly the same cover the empty area. Which piece is it? The pieces may be turned and flipped over.

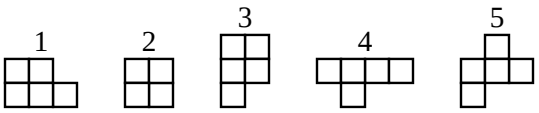


A.  ***B.***  ***C.*** ***D.*** ***E.***

Part C: Each question is worth 5 points

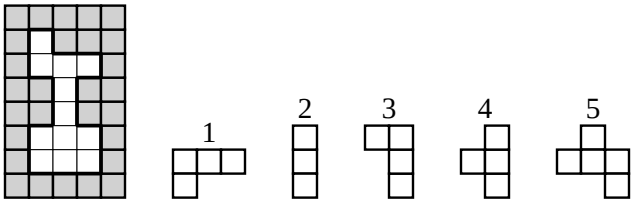
7. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.





A. 1+2+3 ***B.*** 1+3+5 ***C.*** 1+4+5 ***D.*** 2+3+4 ***E.*** 2+3+5

8. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+3+4 ***C.*** 1+3+5 ***D.*** 2+3+5 ***E.*** 2+4+5

Answer Key

1. **D**	2. **D**	3. **C**	4. **B**	5. **A**	6. **D**
7. **A**	8. **E**				

Solutions

1. D — The empty area has 5 squares; all pieces do too, so look at the shape. Turn piece D and it fits exactly; each of the others has one square in a different place. [2003 Q18 · one_piece]

2. D — The empty area has 7 squares; all pieces do too, so look at the shape. Turn piece D and it fits exactly; each of the others has one square in a different place. [2003 Q18 · one_piece]

3. C (2+3) — The empty area has 7 squares. Pieces have 1: 4, 2: 3, 3: 4, 4: 3 squares. Only pieces 2 and 3 also fit together in the right shape. [2003 Q18 · two_pieces]

4. B (1+3) — The empty area has 10 squares. Pieces have 1: 6, 2: 4, 3: 4, 4: 7 squares. Only pieces 1 and 3 also fit together in the right shape. [2003 Q18 · two_pieces]

5. A — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece A fit together into the shape of the hole. [2003 Q18 · two_copies]

6. D — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece D fit together into the shape of the hole. [2003 Q18 · two_copies]

7. A (1+2+3) — The empty area has 14 squares, so the three sizes must add up to 14. Only 1 + 2 + 3 also fit together in the right shape. [2003 Q18 · three_pieces]

8. E (2+4+5) — The empty area has 12 squares, so the three sizes must add up to 12. Only 2 + 4 + 5 also fit together in the right shape. [2003 Q18 · three_pieces]